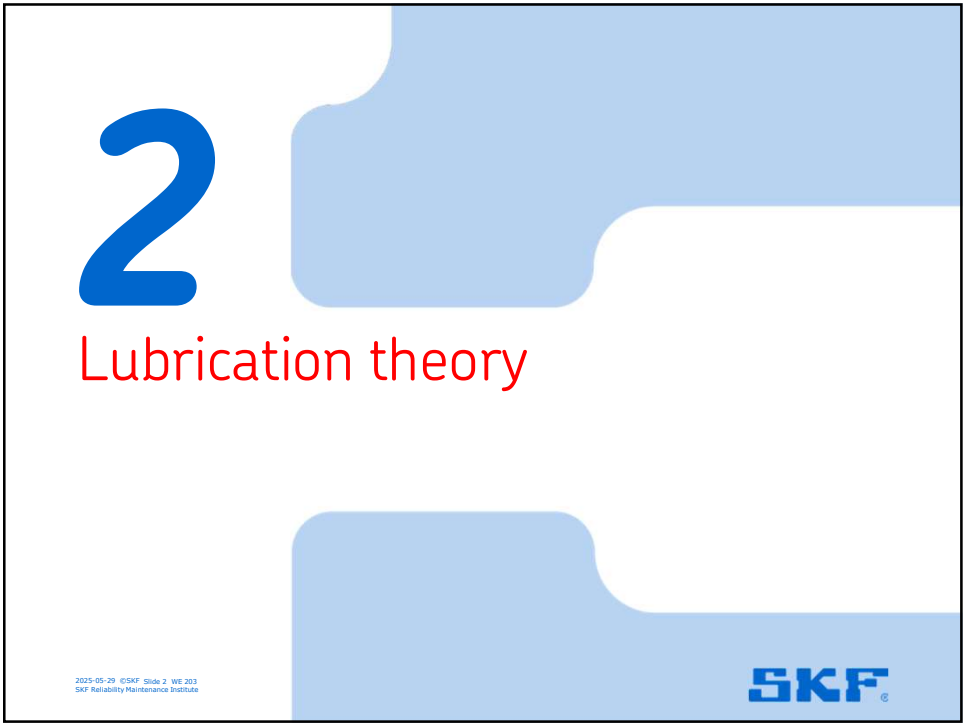


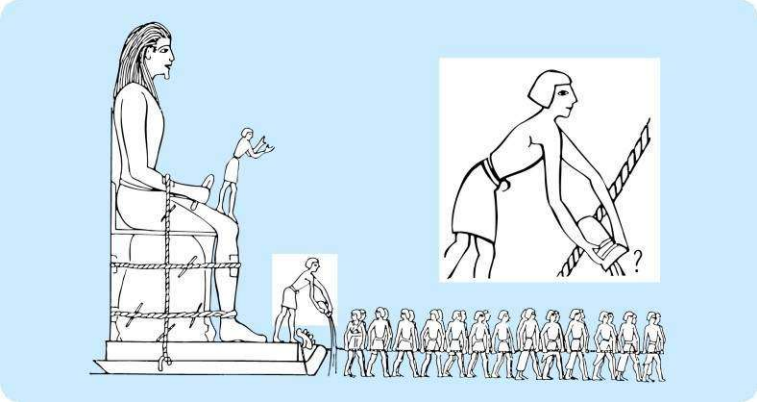


1



2

The fight against friction



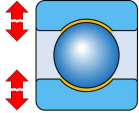
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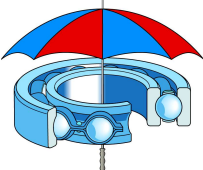
3



Why Lubrication



- **Separate the bearing surfaces**
 - eliminate wear
 - reduce friction

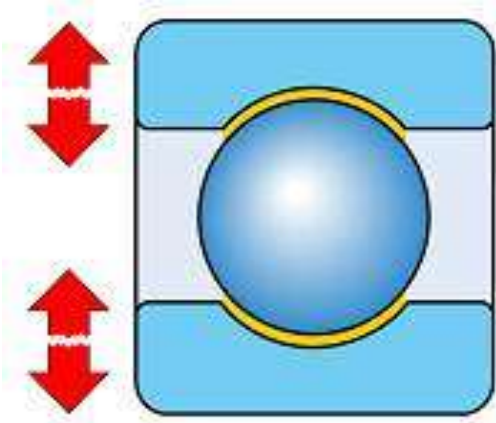


- **Protect the bearing**
 - from corrosion
 - from external impurities

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Why lubrication?

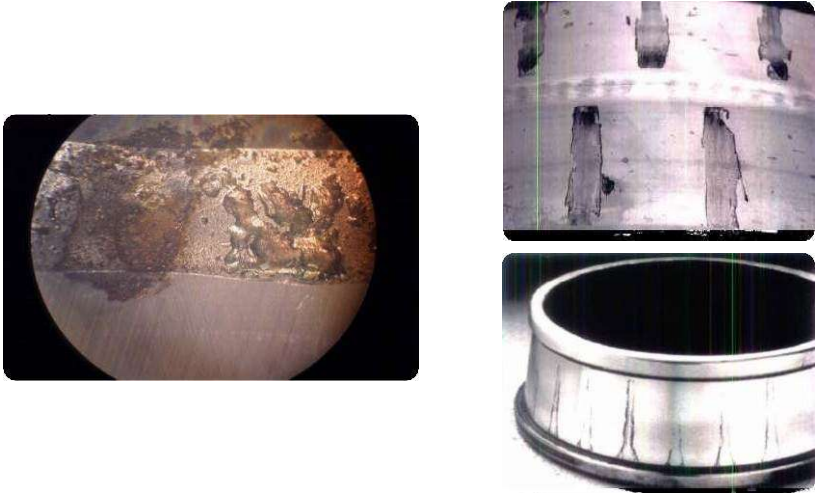
A schematic diagram of a ball bearing. It consists of a central blue sphere (the ball) surrounded by a yellow ring (the inner race). This is housed within a light blue outer ring (the outer race). To the left of the bearing, there are two sets of red double-headed vertical arrows, indicating the bearing's function in allowing relative motion between parts.

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Why lubrication? moisture corrosion

Three photographs illustrating the effects of moisture corrosion on bearing components. The leftmost image is a circular cross-section of a bearing raceway, showing a dark, heavily corroded area on the left side. The top-right image shows a close-up of a bearing's surface with several vertical, dark, pitted areas. The bottom-right image shows a cylindrical bearing component with vertical streaks and a dark, corroded ring at its base.

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Why lubrication? protection

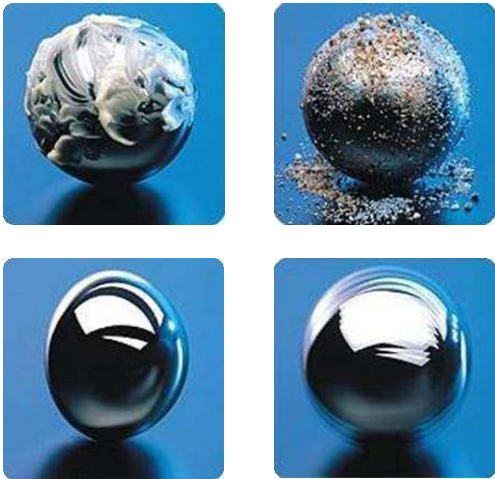


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Prevent over 50% of premature bearing failures...



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Prevent over 50% of premature bearing failures

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How a lubricant works

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Lubricants



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Grease: selection

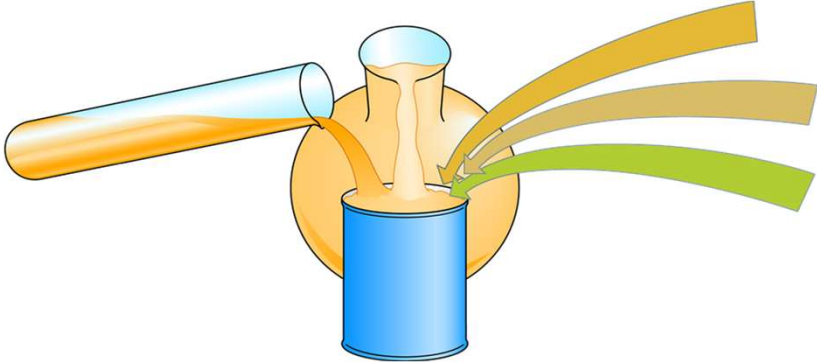


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Grease: structure

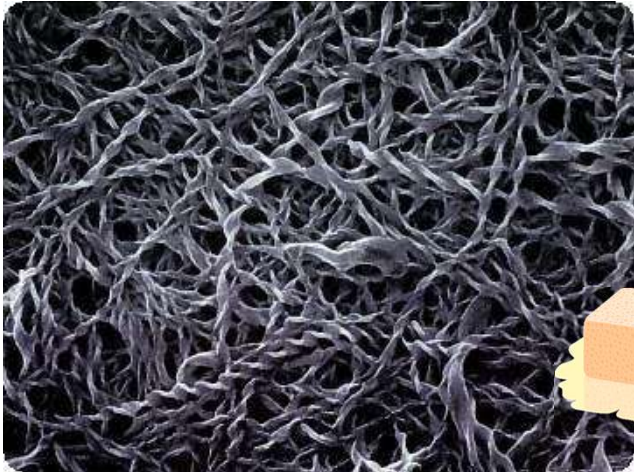
A diagram illustrating the structure of grease. It shows a blue cylindrical container at the bottom. Above it, a yellowish-orange liquid is being poured from a blue tube on the left into a central vertical tube. From the top of this central tube, three thick, curved lines (two yellow and one green) emerge, representing the thickened grease structure.

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Grease structure: thickeners...

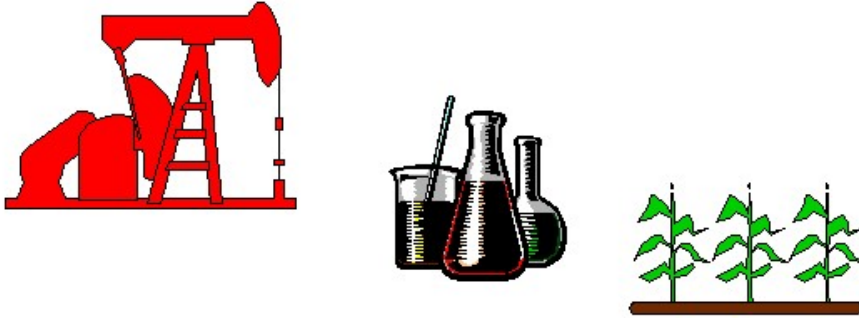
A microscopic image showing a dense, interconnected network of dark, fibrous structures, which are the thickeners in grease. To the right of this image is a small, 3D illustration of a rectangular block of orange grease with a yellowish base.

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(Base) oil



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Oil Additives

Anti-corrosion

AW (anti-wear)

Anti-foam

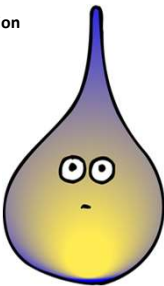
Detergent

Anti-oxidant

Dispersant

EP (extreme pressure)

VI-improvement

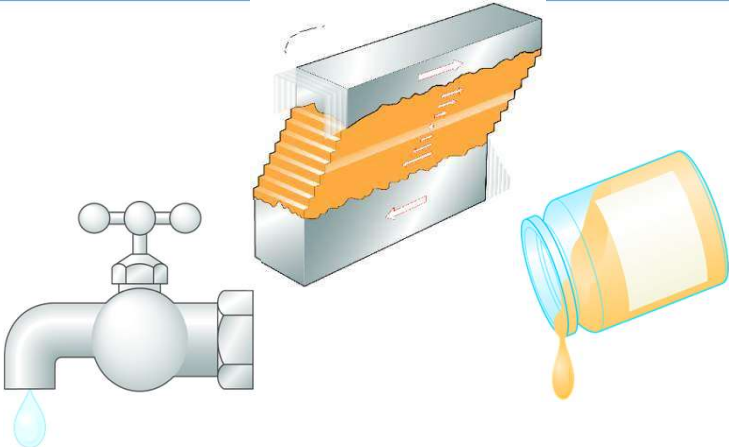


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Viscosity



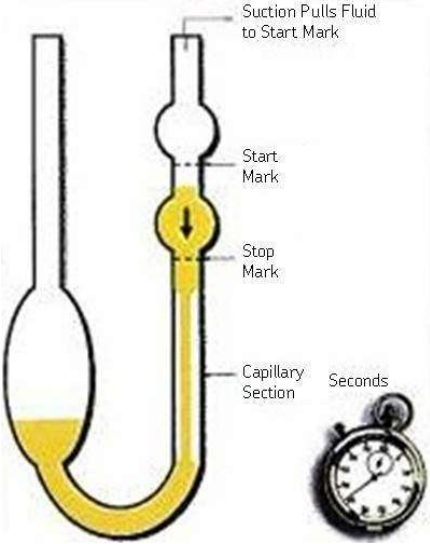
water:
low viscosity
(1 cSt at 40 °C, 105 °F)

honey:
high viscosity
(~1200 cSt at 40 °C, 105 °F)

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(Base) oil viscosity measuring equipment



Suction Pulls Fluid to Start Mark

Start Mark

Stop Mark

Capillary Section

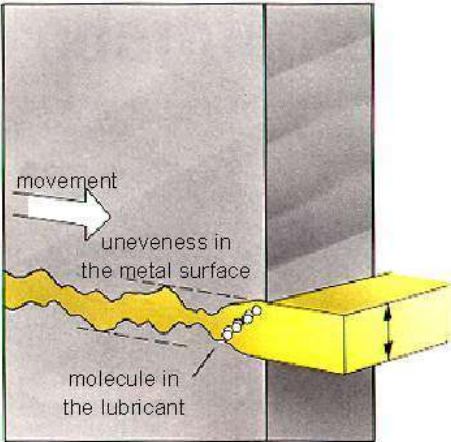
Seconds

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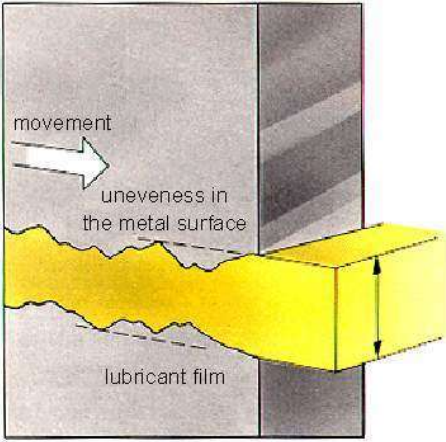
Lubrication regimes

Boundary



The diagram shows two vertical metal surfaces with a wavy line representing an 'unevenness in the metal surface'. A yellow rectangular block is positioned between them. A white arrow labeled 'movement' points to the right. A small circle labeled 'molecule in the lubricant' is shown at the point of contact between the block and the metal surface.

Hydrodynamic



The diagram shows two vertical metal surfaces with a wavy line representing an 'unevenness in the metal surface'. A yellow rectangular block is positioned between them. A white arrow labeled 'movement' points to the right. A yellow layer labeled 'lubricant film' is shown between the block and the metal surface.

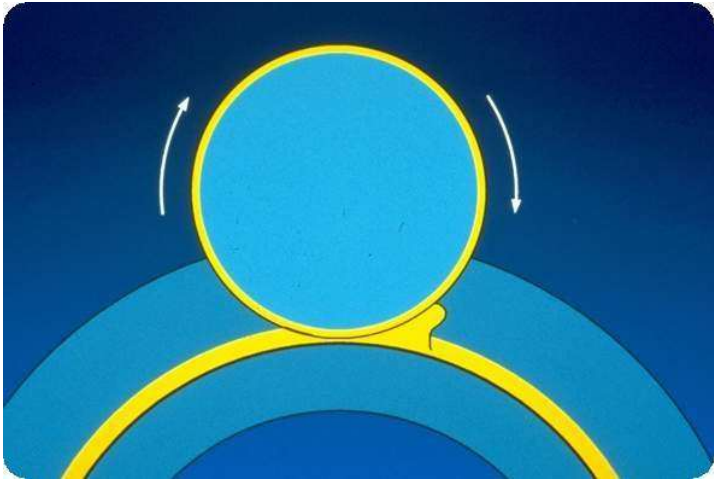
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Lubrication regimes: elasto-hydrodynamic



The diagram shows a cross-section of a bearing and a shaft. The bearing is represented by a blue semi-circle, and the shaft is represented by a yellow semi-circle. A yellow layer representing the lubricant film is shown between them. White curved arrows indicate the direction of rotation for both the bearing and the shaft.

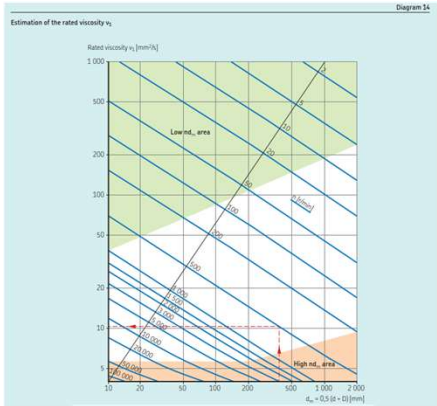
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Rated Viscosity Estimation Chart



RBC 17k Pg 101



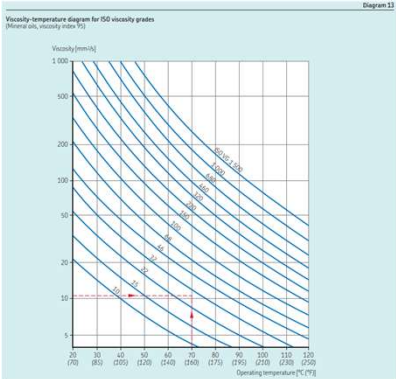
Viscosity Conversion at Ref. Temperature Chart

Table 5

Viscosity classification to ISO 3448

Viscosity grade	Kinematic viscosity limits at 40 °C (105 °F)		
	min.	max.	
	mm²/s		
ISO VG 2	2.2	1.98	2.42
ISO VG 3	3.2	2.88	3.52
ISO VG 5	4.6	4.14	5.06
ISO VG 7	6.8	6.12	7.48
ISO VG 10	10	9.00	11.0
ISO VG 15	15	13.5	16.5
ISO VG 22	22	19.8	24.2
ISO VG 32	32	28.8	35.2
ISO VG 46	46	41.4	50.6
ISO VG 68	68	61.2	74.8
ISO VG 100	100	90.0	110
ISO VG 150	150	135	165
ISO VG 220	220	198	242
ISO VG 320	320	288	352
ISO VG 460	460	414	506
ISO VG 680	680	612	748
ISO VG 1 000	1 000	900	1 100
ISO VG 1 500	1 500	1 350	1 650

RBC 17k Pg 103



RBC 17k Pg 100



Four-step process

- 1) *Select correct consistency*
- 2) *Select necessary additives*
- 3) *Select other properties*
- 4) *Calculate required viscosity*

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Grease consistency: concept

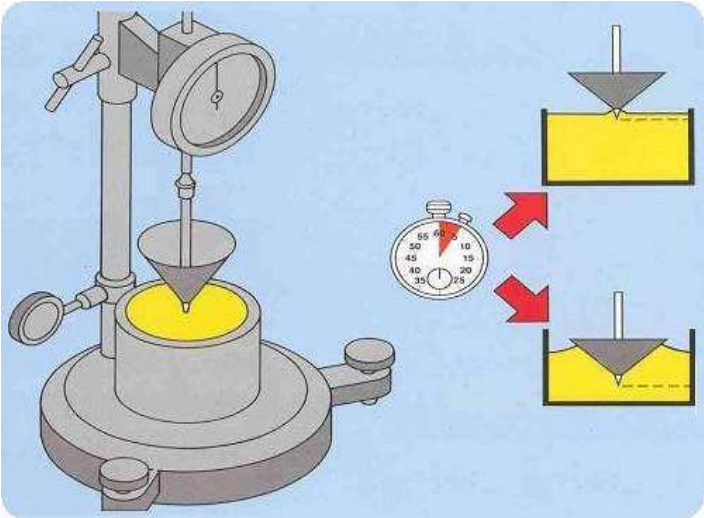


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Grease consistency: consistency determination

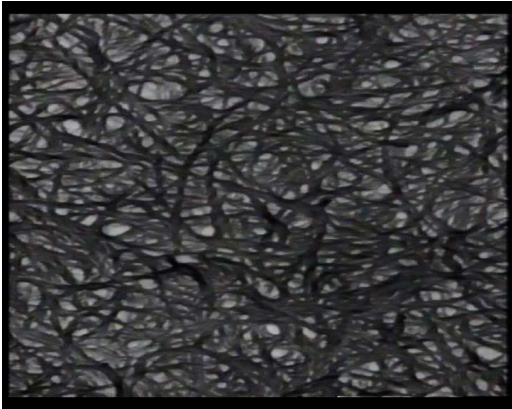


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Penetrometer



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Grease consistency: consistency grades

NLGI Number	ASTM Worked Penetration (tenths of a millimetre)	Appearance at Room Temperature	Use
000	445 – 475	Very Fluid	Gears
00	400 – 430	Fluid	
0	355 – 385	Semi-Fluid	
1	310 – 340	Very Soft	Bearings
2	265 – 295	Soft	
3	220 – 250	Medium Hard	
4	175 – 205	Hard	Sealings
5	130 – 160	Very Hard	
6	85 – 115	Extremely Hard	

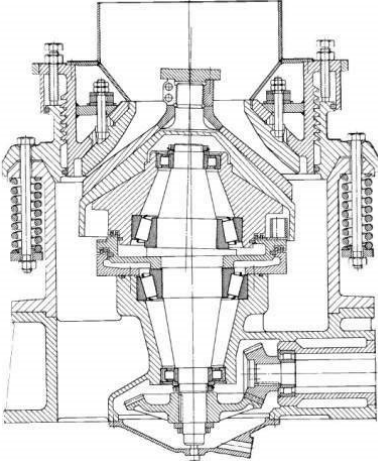
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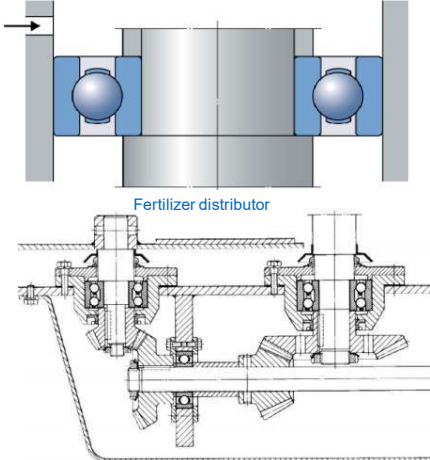
What consistency? vertical shafts NLGI 3

Cone crusher



Cone Crusher

Fertilizer distributor



Fertilizer Distributor

Greases with high consistency grade is beneficial, since more stiffness would help with keeping the grease inside the bearing during operation.

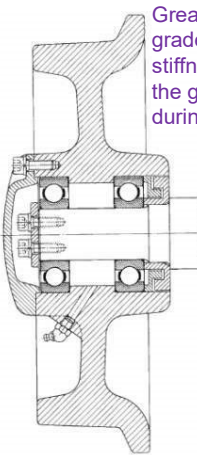
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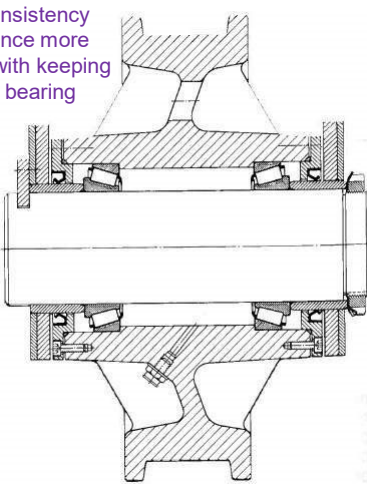
28

What consistency? outer ring rotation NLGI 3

Greases with high consistency grade is beneficial, since more stiffness would help with keeping the grease inside the bearing during operation.



Deep Groove Ball Bearings in Wheel Arrangement



Crane Support Wheels

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What consistency? hot environments (NLGI 3)



Greases with high consistency grade is beneficial, since more stiffness would help with keeping the grease inside the bearing during operation.

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What consistency? cold environments... (NLGI 1)



Greases with low consistency grade is beneficial when continuous ambient temperatures as low as -30°C (-22°F) and when the speed is not high enough

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What consistency? cold environments



Greases with low consistency grade is beneficial when pump grease with high consistency cannot reach the target grease dispense point.

*When pumpability is an issue -
Centralised Lubrication System*

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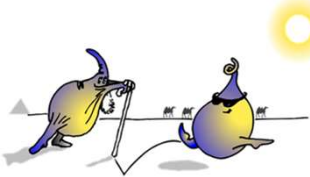


32

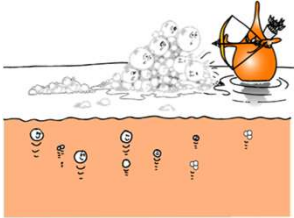
Additives



Anti-Corrosion



Anti-Oxidant



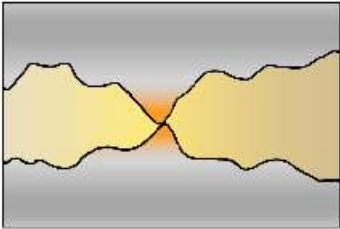
Anti-Foaming

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Additives: extreme pressure – anti-wear



Temperature °C	100	200	300	400	500	600	700	800
Phosphorus	[Yellow bar]							
Chlorine	[Yellow bar]							
Sulphur	[Yellow bar]							

In the high pressure conditions, the viscosity of the base oil on its own is not enough for separation of the mating surfaces.

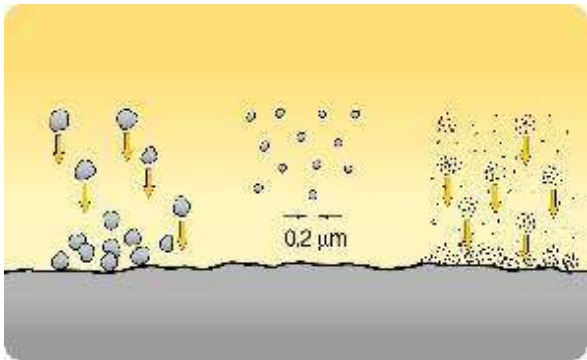
When the tips of the metal surface come into contact with one another, the temperature rises to very high levels at these points, so high that the tips can weld together.

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Additives: solid extreme pressure



For application that boundary-layer lubrication regimes predominate, such as low speed operations, there are conditions in which sufficiently high friction temperatures do not occur, so that no solid film can be formed from the chemical additives then solid lubricating additives such as molybdenum disulphide (MoS2) or graphite would help

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Grease selection: other properties...

LGMT 2

LGMT 3

LGEP 2

LGEM 2

LGEV 2

LGLT 2

LGGB 2

LGWM 1

LGWA 2

LGHB 2

LGHP 2

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Grease selection: other properties

Bearing working conditions	Temp	Speed	Load	Vertical shaft	Fast starting (rotating)	Oscillating movements	Severe vibrations	Shock load or frequent start-up	Low noise	Low friction	Best lubricating properties	Description	Temperature range (°C)		Thickener / base oil	Base oil viscosity (cSt)
													LTL	HTPL		
LGMT 2	M	M	L to H	O	-	-	+	-	-	O	+	General purpose industrial and automotive	-30 °C -22 °F	120 °C 250 °F	Lithium soap / mineral oil	150
LGMT 3	M	M	L to H	+	O	-	+	-	-	O	O	General purpose industrial and automotive	-30 °C -22 °F	120 °C 250 °F	Lithium soap / mineral oil	150
LGEP 2	M	L to H	H	O	-	O	+	+	-	-	+	Extreme pressure	-20 °C -4 °F	150 °C 300 °F	Lithium soap / mineral oil	200
LGFP 2	M	M	L to H	O	-	-	-	-	-	O	+	Food compatible	-20 °C -4 °F	120 °C 250 °F	Aluminium complex / medical white oil	150
LGFB 2	M	M	M to H	O	-	O	+	+	-	-	+	Food compatible, high load	-20 °C -4 °F	120 °C 250 °F	Aluminium complex / PAO and ester	266
LGEM 2	M	VL	H to VH	O	-	+	+	+	-	-	+	High viscosity plus solid lubricants	-20 °C -4 °F	120 °C 250 °F	Lithium soap / mineral oil	500
LGCV 2	M	VL	H to VH	O	-	+	+	+	-	-	+	Extremely high viscosity with solid lubricants	-10 °C 14 °F	120 °C 250 °F	Lithium-calcium soap / mineral oil	1 020
LGTL 2	L to M	M to EH	L	O	-	-	-	O	+	+	O	Low temperature, extremely high speed	-50 °C -58 °F	120 °C 250 °F	Lithium soap / PAO oil	18
LGFL 1	L to M	M to EH	L	-	-	-	-	-	+	+	+	Food compatible, low temperature	-45 °C -49 °F	120 °C 250 °F	Aluminium complex / PAO and ester	30
LGGB 2	L to M	L to M	M to H	O	-	+	+	+	-	O	O	Green biodegradable, low toxicity	-40 °C -40 °F	90 °C 175 °F	Lithium-calcium soap / synthetic ester oil	150
LGWM 1	L to M	L to M	H	-	-	+	-	+	-	-	+	Extreme pressure, low temperature	-30 °C -22 °F	150 °C 300 °F	Lithium soap / mineral oil	200
LGWA 2	M to H	L to M	H	O	O	O	O	+	-	O	+	Wide temperature (°F), extreme pressure	-30 °C -22 °F	140 °C 284 °F	Lithium complex soap / mineral oil	185
LGHB 2	M to H	VL to M	H to VH	O	+	+	+	+	-	-	+	EP high viscosity, high temperature (°F)	-20 °C -4 °F	150 °C 300 °F	Complex calcium sulfonate / mineral oil	400
LGHF 2	M to H	M to H	L to M	O	-	-	+	O	+	O	+	High performance polyurea grease	-40 °C -40 °F	150 °C 300 °F	Di-urea / mineral oil	96
LGTE 2	VH	L to M	H to VH	O	+	+	O	O	-	-	O	Extreme temperature	-40 °C -40 °F	260 °C 500 °F	PTFE / synthetic (fluorinated polyether)	400

(°F) LTL = Low temperature limit
(°F) HTPL = High temperature performance limit
(°F) HTPL = High temperature performance limit
(°F) HTPL = High temperature performance limit

(*) LGMT 2 can withstand peak temperatures of 120 °C / 250 °F
(*) LGMT 3 can withstand peak temperatures of 120 °C / 250 °F
(*) LGMT 3 can withstand peak temperatures of 120 °C / 250 °F

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SKF Grease Selection Chart

Bearing working conditions	Temp	Speed	Load	Vertical shaft	Fast starting (rotating)	Oscillating movements	Severe vibrations	Shock load or frequent start-up	Low noise	Low friction	Best lubricating properties	Description	Temperature range (°C)		Thickener / base oil	Base oil viscosity (cSt)
													LTL	HTPL		
LGMT 2	M	M	L to H	O	-	-	+	-	-	O	+	General purpose industrial and automotive	-30 °C -22 °F	120 °C 250 °F	Lithium soap / mineral oil	150
LGMT 3	M	M	L to H	+	O	-	+	-	-	O	O	General purpose industrial and automotive	-30 °C -22 °F	120 °C 250 °F	Lithium soap / mineral oil	150
LGEP 2	M	L to H	H	O	-	O	+	+	-	-	+	Extreme pressure	-20 °C -4 °F	150 °C 300 °F	Lithium soap / mineral oil	200
LGFP 2	M	M	L to H	O	-	-	-	-	-	O	+	Food compatible	-20 °C -4 °F	120 °C 250 °F	Aluminium complex / medical white oil	150
LGFB 2	M	M	M to H	O	-	O	+	+	-	-	+	Food compatible, high load	-20 °C -4 °F	120 °C 250 °F	Aluminium complex / PAO and ester	266
LGEM 2	M	VL	H to VH	O	-	+	+	+	-	-	+	High viscosity plus solid lubricants	-20 °C -4 °F	120 °C 250 °F	Lithium soap / mineral oil	500
LGCV 2	M	VL	H to VH	O	-	+	+	+	-	-	+	Extremely high viscosity with solid lubricants	-10 °C 14 °F	120 °C 250 °F	Lithium-calcium soap / mineral oil	1 020
LGTL 2	L to M	M to EH	L	O	-	-	-	O	+	+	O	Low temperature, extremely high speed	-50 °C -58 °F	120 °C 250 °F	Lithium soap / PAO oil	18
LGFL 1	L to M	M to EH	L	-	-	-	-	-	+	+	+	Food compatible, low temperature	-45 °C -49 °F	120 °C 250 °F	Aluminium complex / PAO and ester	30
LGGB 2	L to M	L to M	M to H	O	-	+	+	+	-	O	O	Green biodegradable, low toxicity	-40 °C -40 °F	90 °C 175 °F	Lithium-calcium soap / synthetic ester oil	150
LGWM 1	L to M	L to M	H	-	-	+	-	+	-	-	+	Extreme pressure, low temperature	-30 °C -22 °F	150 °C 300 °F	Lithium soap / mineral oil	200
LGWA 2	M to H	L to M	H	O	O	O	O	+	-	O	+	Wide temperature (°F), extreme pressure	-30 °C -22 °F	140 °C 284 °F	Lithium complex soap / mineral oil	185
LGHB 2	M to H	VL to M	H to VH	O	+	+	+	+	-	-	+	EP high viscosity, high temperature (°F)	-20 °C -4 °F	150 °C 300 °F	Complex calcium sulfonate / mineral oil	400
LGHF 2	M to H	M to H	L to M	O	-	-	+	O	+	O	+	High performance polyurea grease	-40 °C -40 °F	150 °C 300 °F	Di-urea / mineral oil	96
LGTE 2	VH	L to M	H to VH	O	+	+	O	O	-	-	O	Extreme temperature	-40 °C -40 °F	260 °C 500 °F	PTFE / synthetic (fluorinated polyether)	400

(°F) LTL = Low temperature limit
(°F) HTPL = High temperature performance limit
(°F) HTPL = High temperature performance limit
(°F) HTPL = High temperature performance limit

(*) LGMT 2 can withstand peak temperatures of 120 °C / 250 °F
(*) LGMT 3 can withstand peak temperatures of 120 °C / 250 °F
(*) LGMT 3 can withstand peak temperatures of 120 °C / 250 °F

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Selection of a SKF grease

[LubeSelect for SKF greases | SKF](#)



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What viscosity?...



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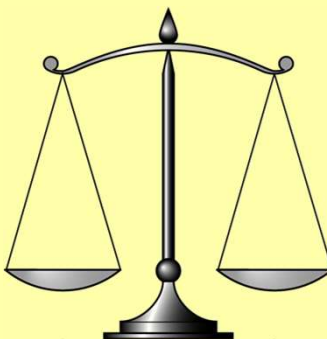


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Considerations for Oil Viscosity

Low viscosity

- + low friction
- thin oil film



High viscosity

- + thick oil film
- high friction

Find the right balance

Base oil viscosity changes with temperature (like with honey)

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What viscosity?

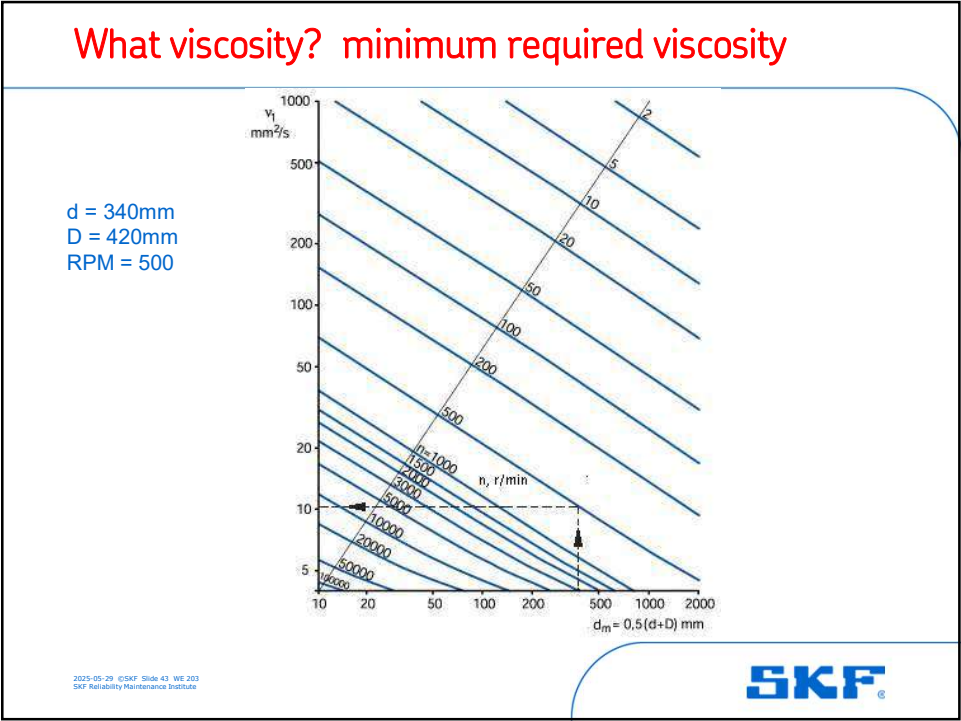




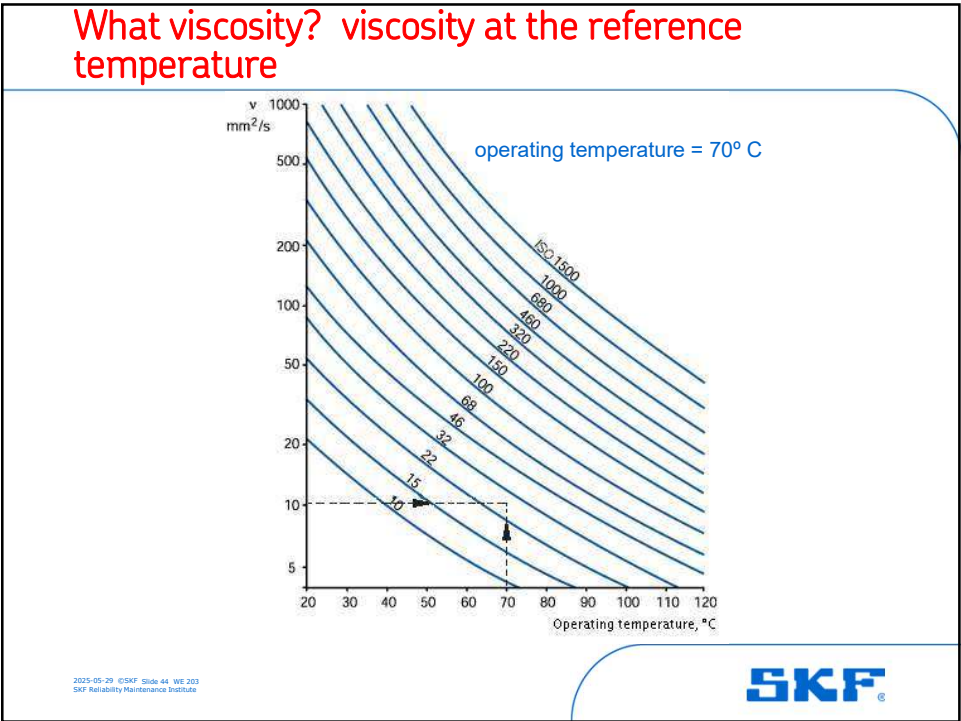
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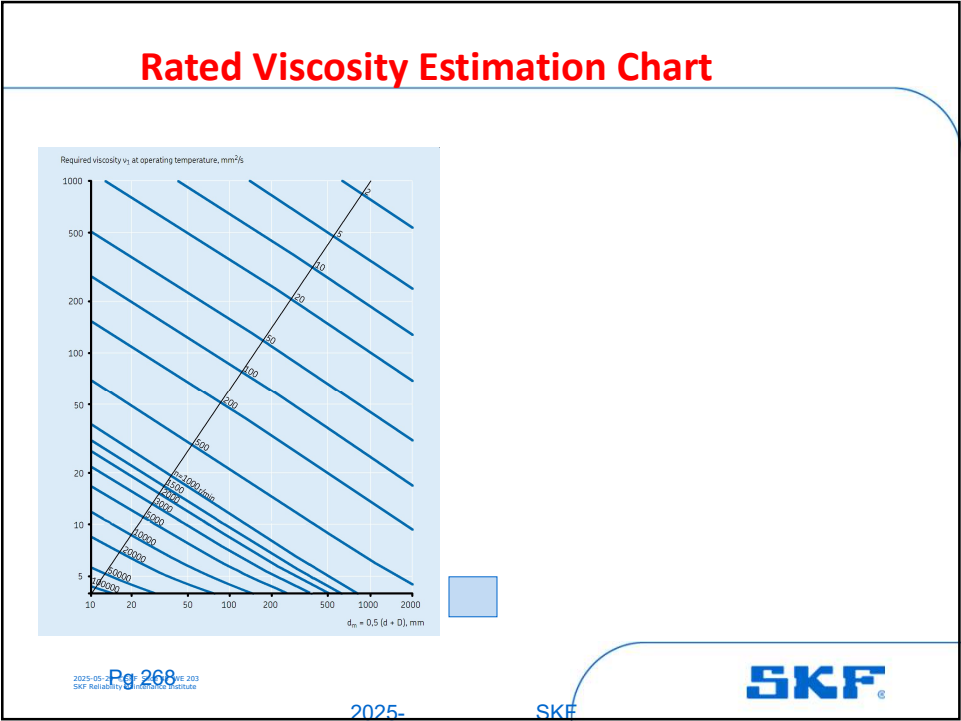
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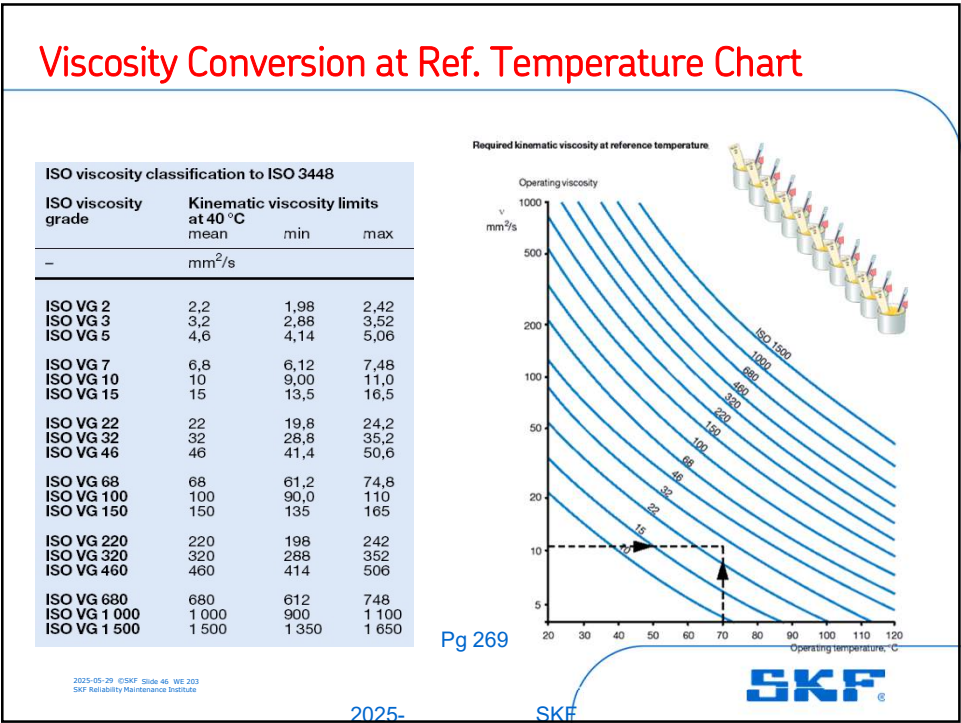
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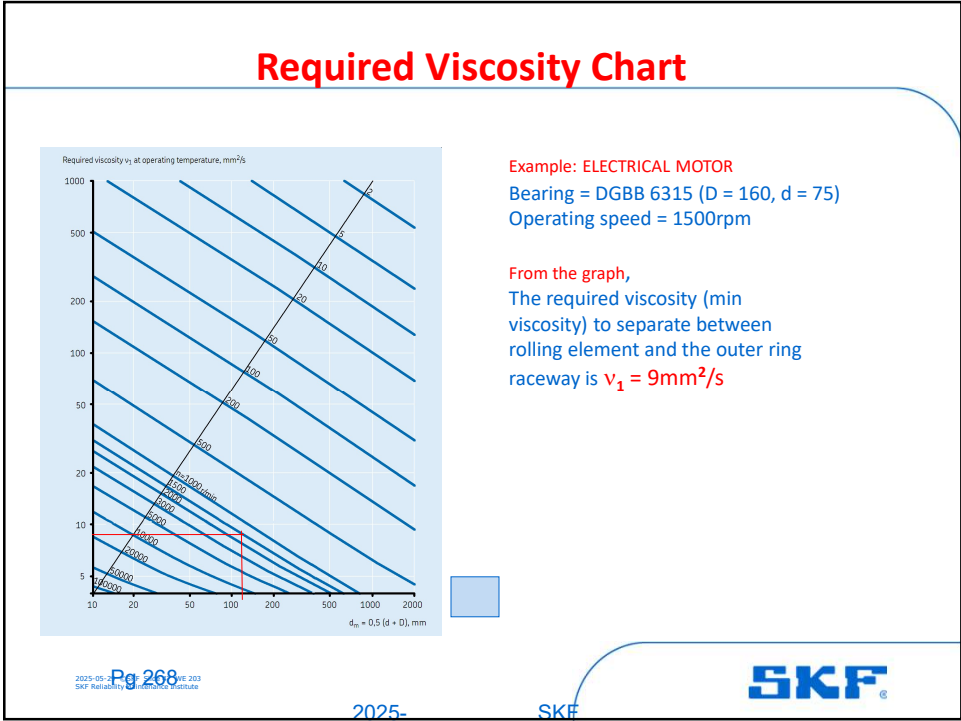
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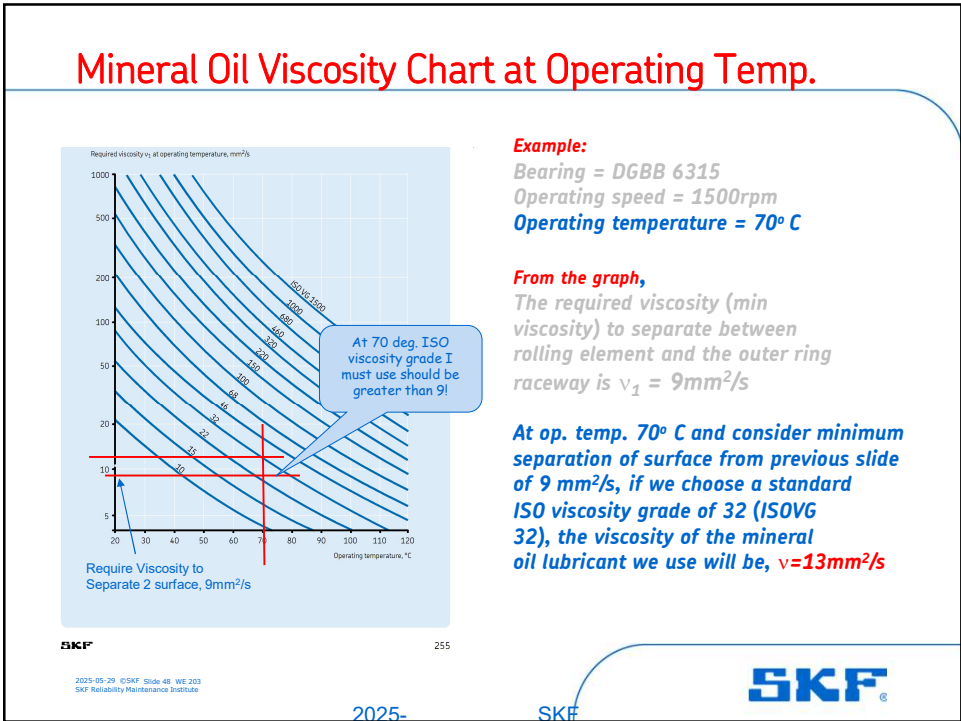
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46



47



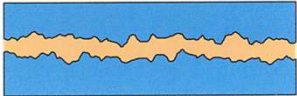
48

The oil film effect on life

$$\kappa = \frac{v}{v_1}$$

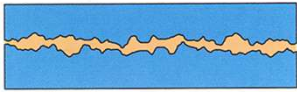
v = Operating viscosity
 v_1 = Required viscosity

Full film lubrication



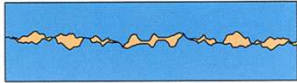
$\kappa > 4$

Mixed lubrication



$\kappa = 1 - 4$

Boundary lubrication



$\kappa < 1$

The film thickness depends on:

- Bearing size
- Rotational speed
- Oil viscosity

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Viscosity Charts (example 1)

Electrical Motor-6315 d = 75mm D=160mm B= 37mm
Speed, n = 1500rpm
 $d_m = (d + D)/2 = 117.5\text{mm}$

- Page 254, Required Viscosity, $v_1 = 9\text{mm}^2/\text{s}$ @ 70°C
- Page 255, can use oil with ISO VG 32 or higher

At 70°C, ISO VG 32 gave $v = 13\text{mm}^2/\text{s}$

- Kappa ratio, $\kappa = v/v_1 = 13/9 = 1.4$
- Interval of re-lubrication

$A = n \times d_m = 1500 \times 117.5 = 176\,250$
 $A \times b_f = 176\,250 \times 1 = 176\,250$ b_f – GC Pg 239

From diagram if C/P = 8 then $t_f = 6500\text{hrs} = 270.8\text{days} = 9\text{months}$
If C/P= 4 then $t_f = 2500\text{hrs} = 104\text{days} = 3.5\text{months}$

Quantity, $G = 0.005 \times D \times B = 0.005 \times 160 \times 75$
 $= 29.6\text{grams}$

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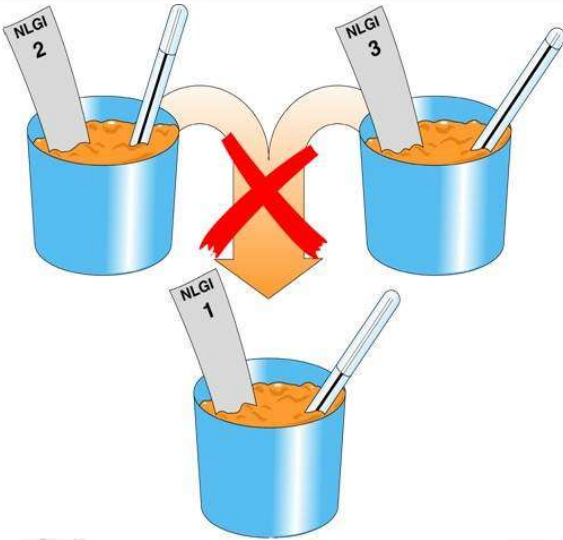
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Compatibility of greases

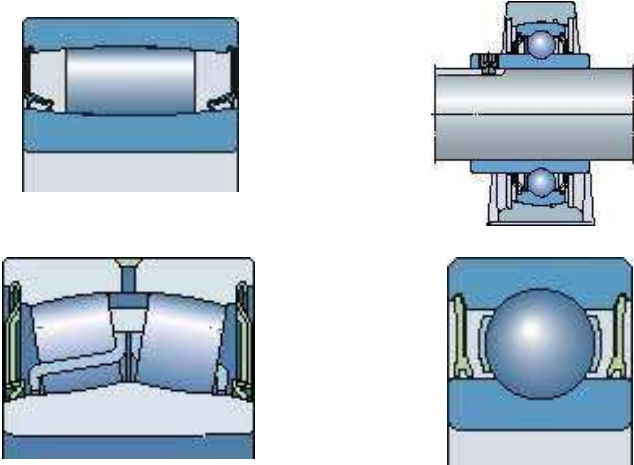


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Grease charge in the closed bearings



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Deep groove ball bearings with shields or seals...

Grease charge

Diameter Series	Outside diameter, mm		30 < D ≤ 62	D > 62
	D ≤ 30 d < 10	D ≤ 30 d ≥ 10		
8, 9	LHT23	LT10	MT47	MT33
0, 1, 2, 3	MT47	MT47	MT47	MT33

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Deep groove ball bearings with shields or seals

Standard and special grease charge

Technical specifications	Standard greases			Special greases				
	LT10	MT47	MT33	GJN	HT22	LT20	GWB	LHT23
Thickener	Lithium soap	Lithium soap	Lithium soap	Polyurea soap	Lithium complex soap	Lithium soap	Polyurea soap	Lithium soap
Base oil type	Diester oil	Mineral oil	Mineral oil	Mineral oil	Mineral oil	Diester oil	Ester oil	Ester oil
NLGI Consistency Class	2	2	3	2	3	2	2-3	2
Operating Temperature, °C	-50 to 90	-30 to 110	-30 to 120	-30 to 150	-20 to 140	-55 to 110	-40 to 160	-50 to 140
Base oil viscosity, mm²/s								
at 40° C	12	70	74	115	110	15	70	26
at 100° C	3,3	7,3	8,5	12,2	13	3,7	9,4	5,1

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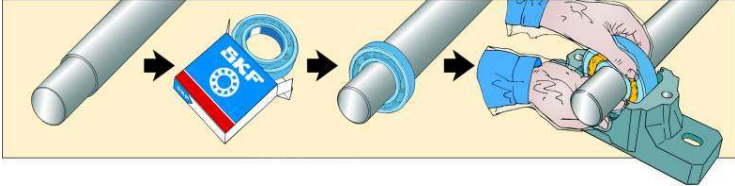
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The right time

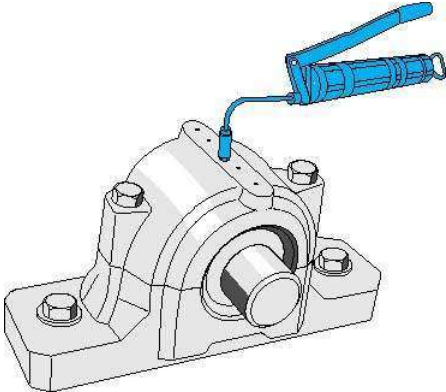


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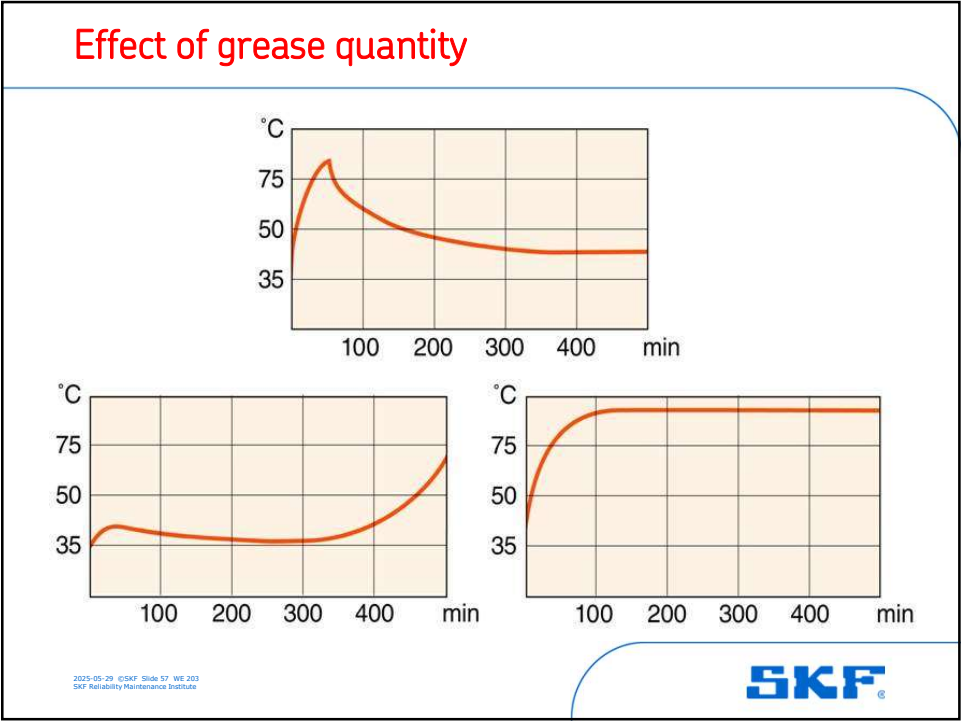
How much grease?



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Grease quantity for the standard housings



Grease level markings
To avoid over-filling, markings at each corner inside the housing base indicate the proper grease level for an initial 20 or 40% fill (fig. 8).



Housing Size	Grease quantities		Housing Size	Grease quantities	
	First fill ¹⁾	Re-lub-rication		First fill ¹⁾	Re-lub-rication
-	g		-	g	
SNL 205	25	5	SNL 511-609	100	15
SNL 206-305	40	5	SNL 512-610	150	15
SNL 207	50	10	SNL 513-611	180	20
SNL 208-307	60	10	SNL 515-612	230	20
SNL 209	65	10	SNL 516-613	280	25
SNL 210	75	10	SNL 517	330	25
SNL 211	100	15	SNL 518-615	430	40
SNL 212	150	15	SNL 519-616	480	50
SNL 213	180	20	SNL 520-617	630	55
SNL 215	230	20	SNL 522-619	850	70
SNL 216	280	25	SNL 524-620	1 000	80
SNL 217	330	25	SNL 526	1 100	95
SNL 218	430	40	SNL 528	1 400	110
SNL 505	25	5	SNL 530	1 700	130
SNL 506-605	40	5	SNL 532	2 000	150
SNL 507-606	50	10			
SNL 508-607	60	10			
SNL 509	65	10			
SNL 510-608	75	10			

¹⁾ Fills approximately 40% of the free space in the housing

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Replenishment: quantity

1

2

- 40% when replenishing is made from the side of the bearing (fig 1)
- 20% when replenishing is made through the annular groove and lubrication holes in the bearing outer or inner ring (fig 2)

Suitable quantities for replenishment from the side of a bearing can be obtained from:
 $G_p = 0.005 D B$
and for replenishment through the bearing outer or inner ring from:
 $G_p = 0.002 D B$
where:
 G_p = grease quantity to be added when replenishing, g
 D = bearing outside diameter, mm
 B = total bearing width (for thrust bearings, use total height H), mm

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Replenishment: time

Re-lubrication intervals at operating temperatures of 70 °C

Bearing type ¹⁾	Bearing factor f_1	Recommended limits for speed factor A and load ratio C/P		
		$C/P \geq 15$	$C/P = 8$	$C/P = 4$
Deep groove ball bearings	1	500 000	400 000	300 000
Angular contact ball bearings	1	500 000	400 000	300 000
Self-aligning ball bearings	1	500 000	400 000	300 000
Cylindrical roller bearings				
- full complement bearing	1.5	400 000	300 000	150 000
- with 1/2 full complement bearing	2	300 000	200 000	100 000
- with 1/4 full complement bearing	4	250 000	150 000	60 000
- full complement ²⁾	4	NA ³⁾	NA ³⁾	20 000
Taper roller bearings	2	350 000	200 000	200 000
Spherical roller bearings				
- when load ratio $F_R/F_N \leq 0.5$ and $d_{pw} \leq 600$ mm	2	350 000	200 000	100 000
- when load ratio $F_R/F_N \leq 0.5$ and $d_{pw} > 600$ mm	2	250 000	150 000	80 000
- when load ratio $F_R/F_N > 0.5$ and $d_{pw} \leq 600$ mm	2	200 000	100 000	60 000
- when load ratio $F_R/F_N > 0.5$ and $d_{pw} > 600$ mm	2	150 000	80 000	40 000
- all series	6	150 000	80 000 ⁴⁾	30 000 ⁴⁾
CARB toroidal roller bearings				
- with cage	2	350 000	200 000	100 000
- without cage, full complement ⁵⁾	4	NA ³⁾	NA ³⁾	20 000
Thrust ball bearings	2	200 000	150 000	100 000
Cylindrical roller thrust bearings	10	100 000	60 000	30 000
Spherical roller thrust bearings	4	200 000	170 000	150 000

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Oil Lubrication Selection Procedure

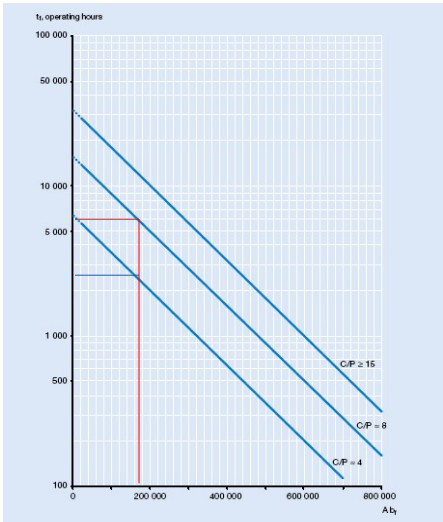
Electrical Motor-6315 d = 75mm D=160mm B= 37mm
Speed, n = 1500rpm
 $d_m = (d + D)/2 = 117.5\text{mm}$
• Page 268, Required Viscosity, $v_1 = 9\text{mm}^2/\text{s}$ @ 70°C
• Page 269, can use oil with ISO VG 32 or higher
At 70°C, ISO VG 32 gave $v = 13\text{mm}^2/\text{s}$
• Kappa ratio, $\kappa = v/v_1 = 13/9 = 1.4$
• Interval of re-lubrication
 $A = n \times d_m = 1500 \times 117.5 = 176\,250$
 $A \times b_f = 176\,250 \times 1 = 176\,250$
From diagram if C/P = 8 then $t_f = 6500\text{hrs} = 270.8\text{days} = 9\text{months}$
If C/P = 4 then $t_f = 2500\text{hrs} = 104\text{days} = 3.5\text{months}$
Quantity, $G = 0.005 \times D \times B = 0.005 \times 160 \times 37$
 $= 29.6\text{grams}$

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Re-lubrication interval at operating temperature of 70°C



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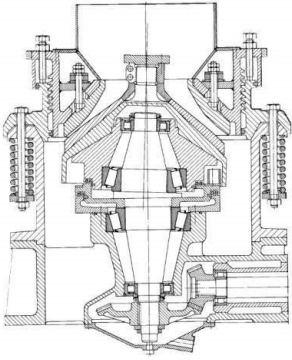
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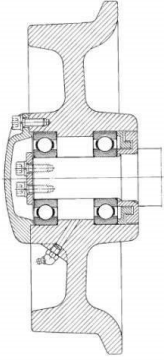
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Adjustments of re-lubrication intervals



Vertical Shaft
Re-lubrication halved



Outer Ring Rotation
Used D instead of d_m
For speed factor a calculation



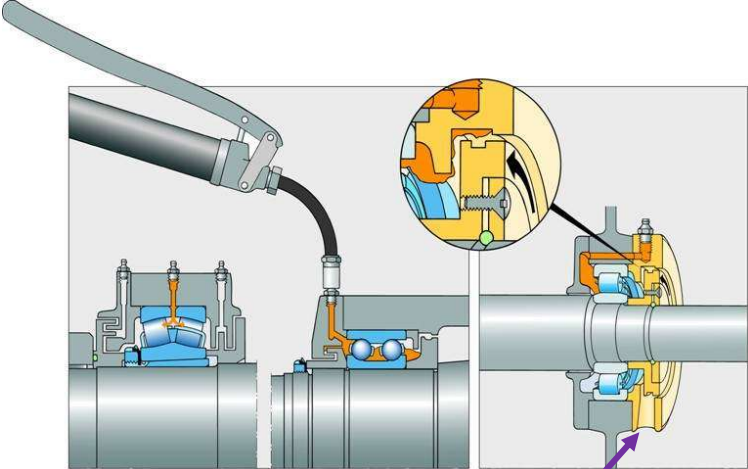
Cylindrical Roller Bearings with Cage J, MA, ML
Re-lubrication halved

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Preventing excessive grease

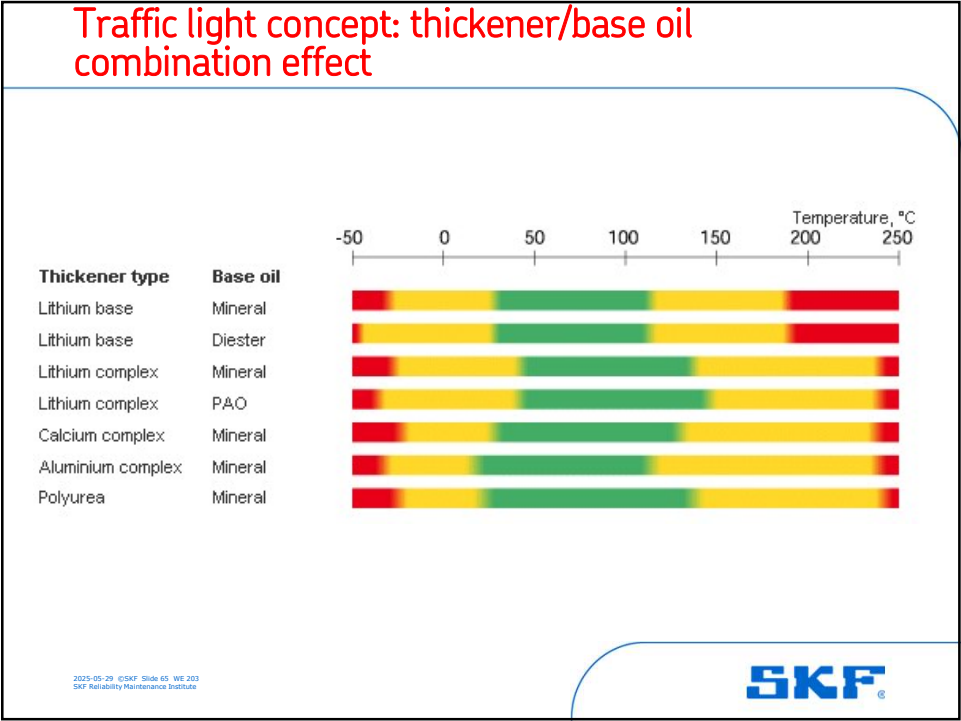


Grease escape valve
For excessive grease

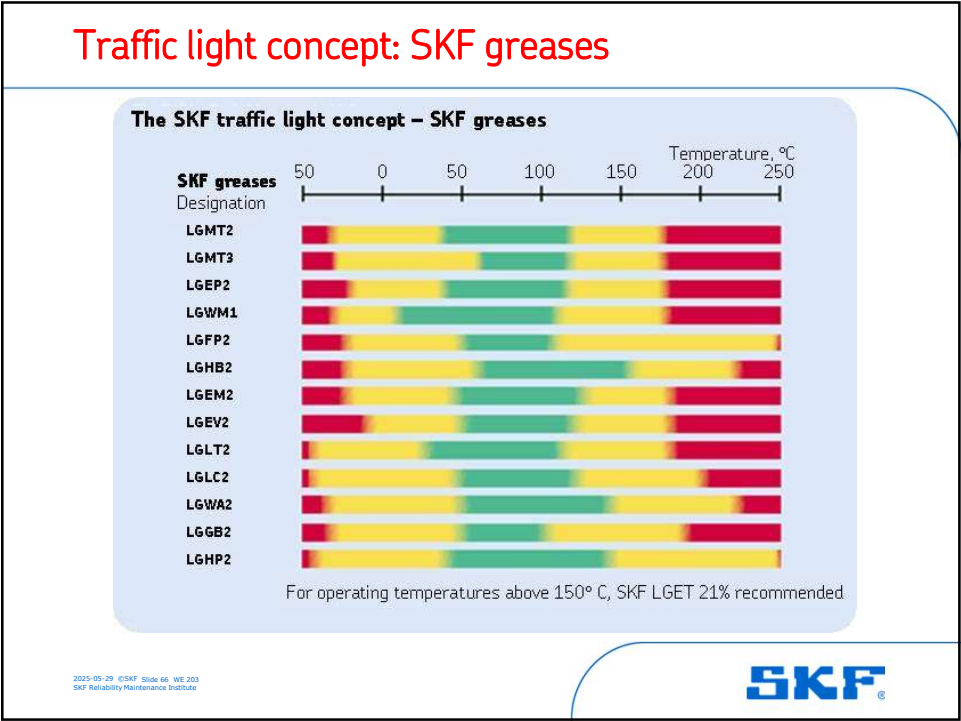
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Continuous lubrication



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Continuous lubrication: single point automatic lubricators

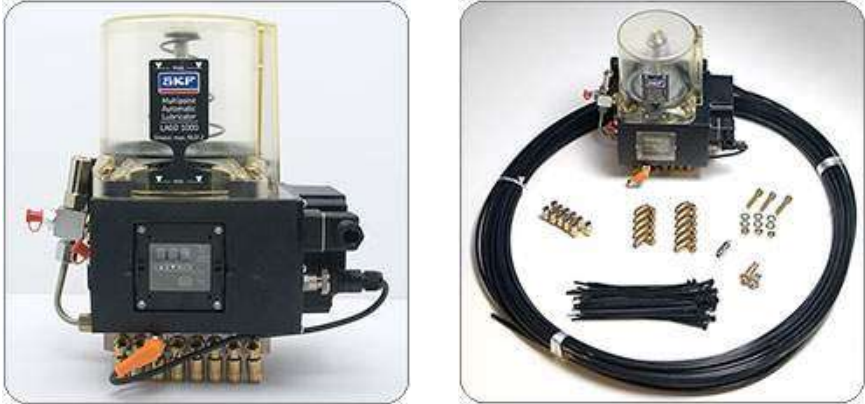


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Multipoint lubricator: System LAGD 1000

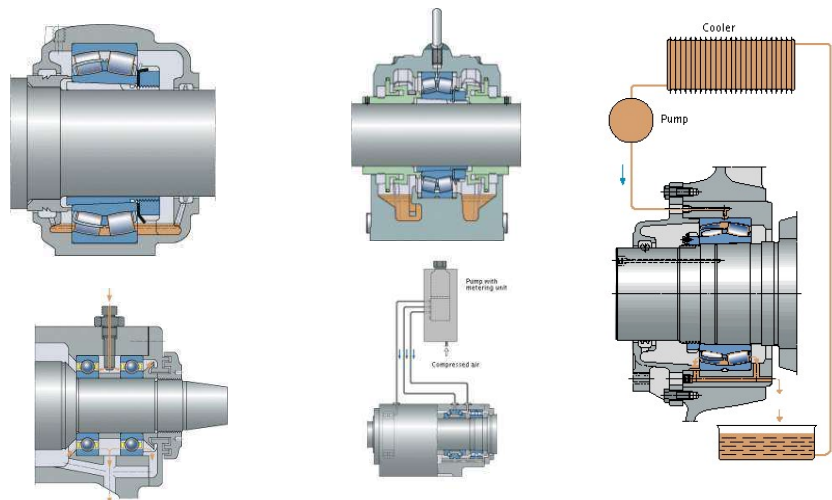


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Oil lubrication



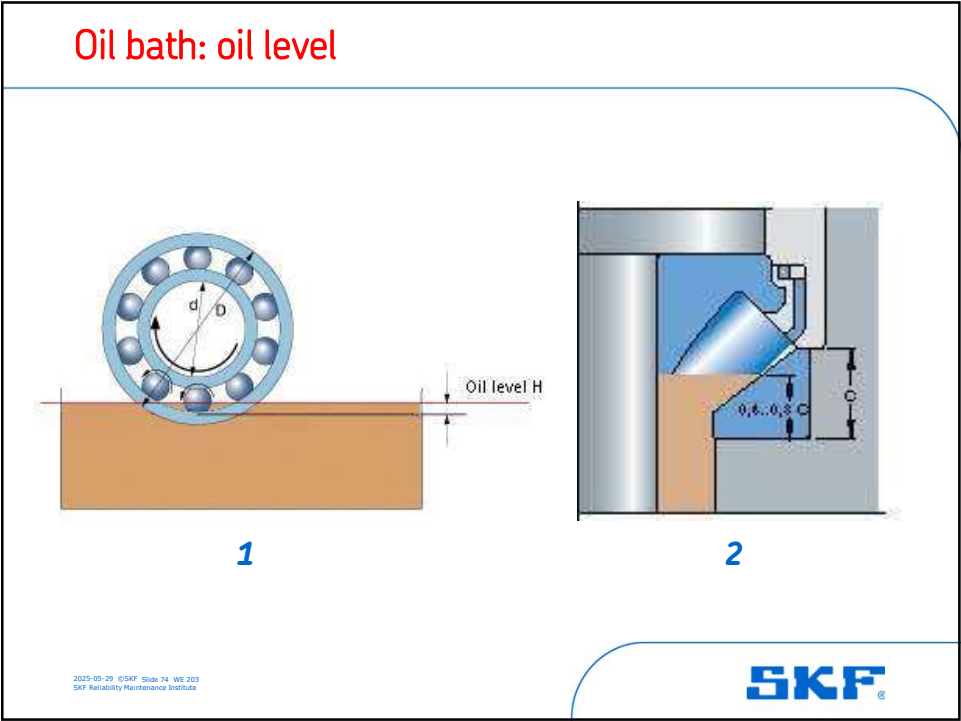
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Oil bath: oil leveller

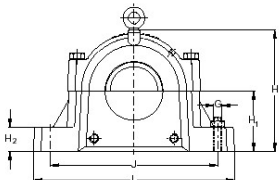


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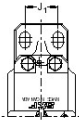


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Oil bath: static oil level in pillow block housings



Pillow block housing series SAFD

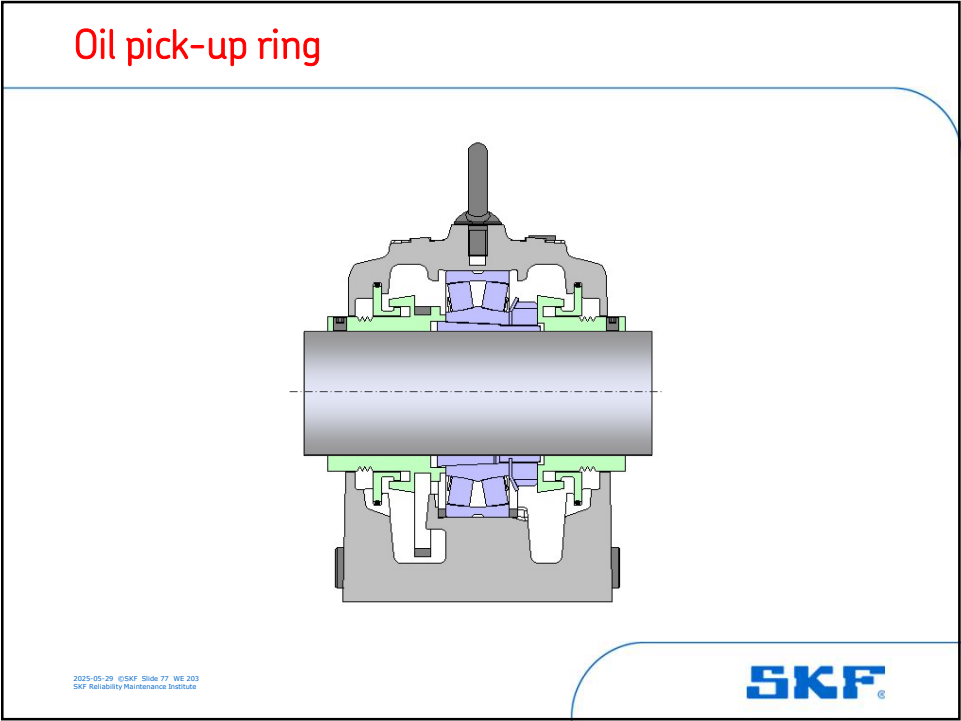


Housing No.	Dimensions							Static Oil Level					Bolts (req'd)		End Plug	
	A	A ₁	A ₂	A ₃	H	H ₁	H ₂	H ₃	L	J	J _{max}	J _{min}	J ₁	G		S
	in/mm															
SAFD 507	3 ³ / ₁₆	2	3 ¹ / ₄	-	3 ¹³ / ₁₆	2	¹³ / ₁₆	³¹ / ₃₂	7 ¹ / ₂	6 ¹ / ₈	5 ⁵ / ₈	-	(2) ¹ / ₂	*	EPR 2	
	81	51	83	-	97	50.8	21	25	191	156	143	-	(2) 12			
SAFD 509	3 ⁷ / ₁₆	2 ³ / ₈	3 ⁵ / ₈	-	4 ⁹ / ₈	2 ¹ / ₄	¹³ / ₁₆	³¹ / ₃₂	8 ¹ / ₄	7	6 ¹ / ₄	-	(2) ¹ / ₂	*	EPR 3	
	87	60	92	-	111	57.15	21	25	210	178	159	-	(2) 12			

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Circulating oil

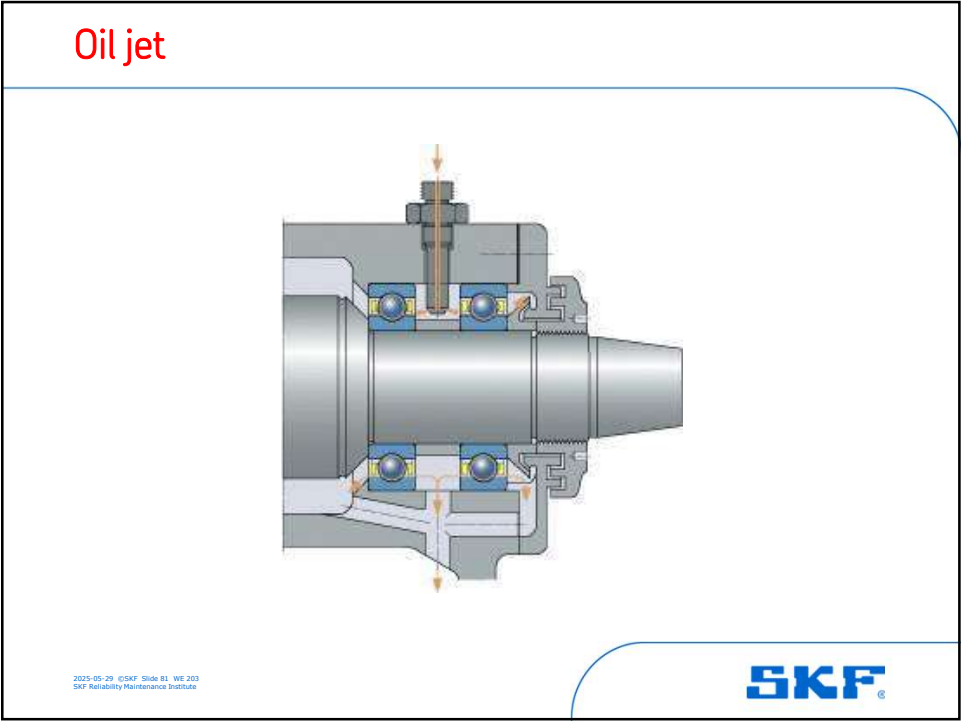
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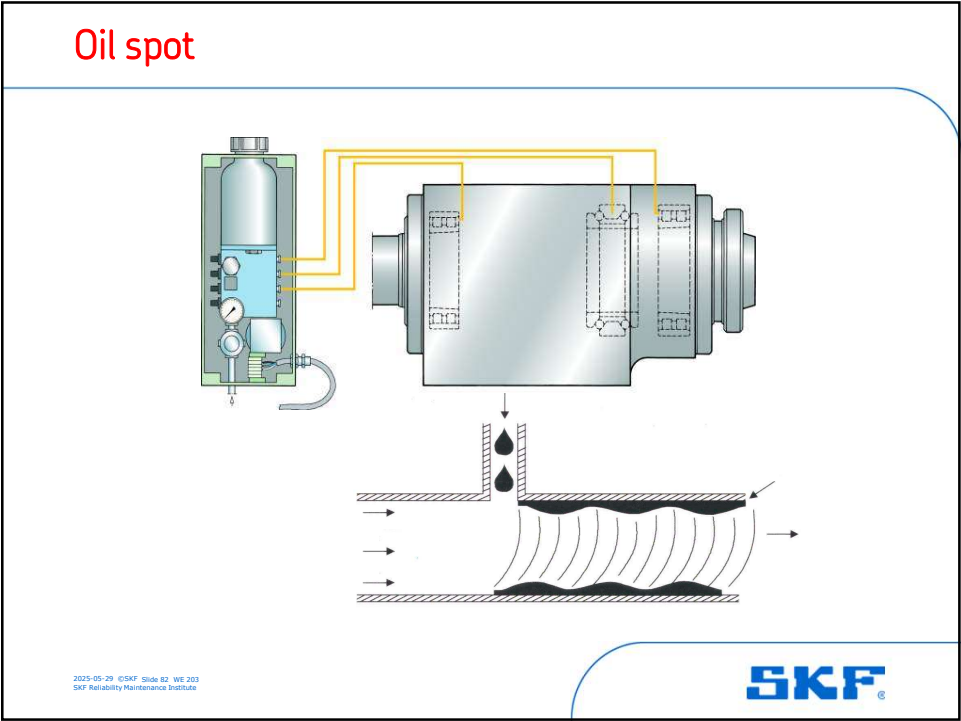
Circulating oil: wet sump/dry sump

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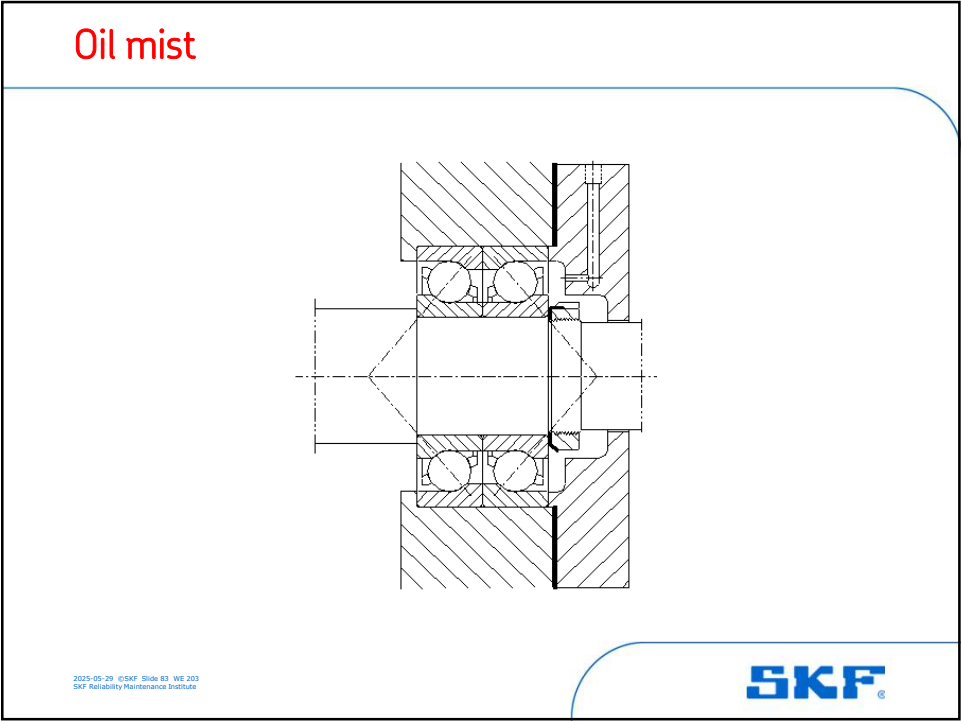
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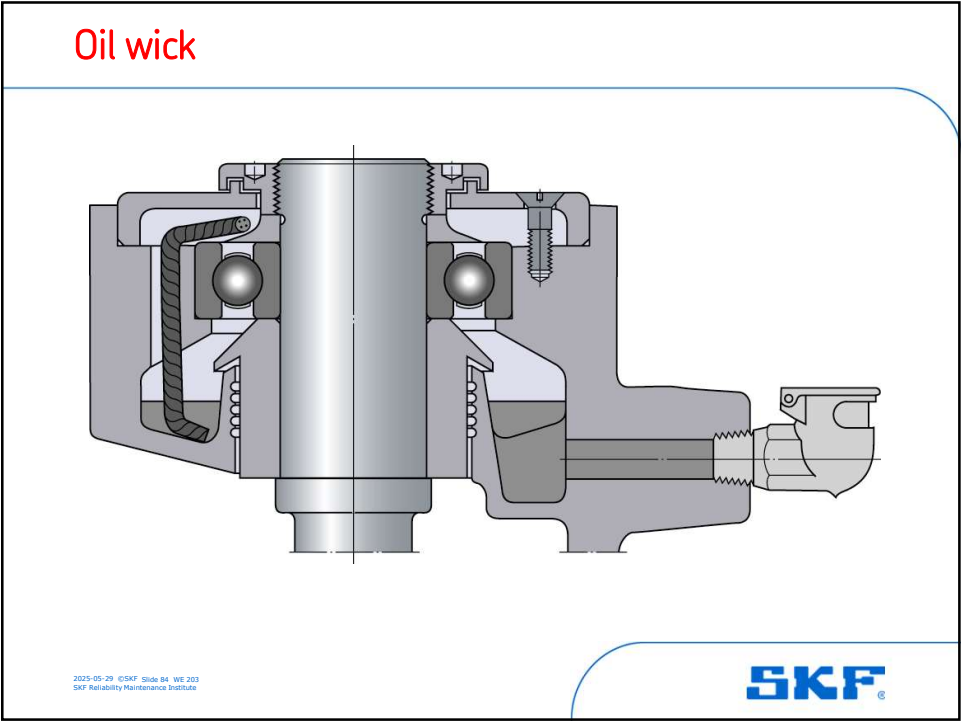
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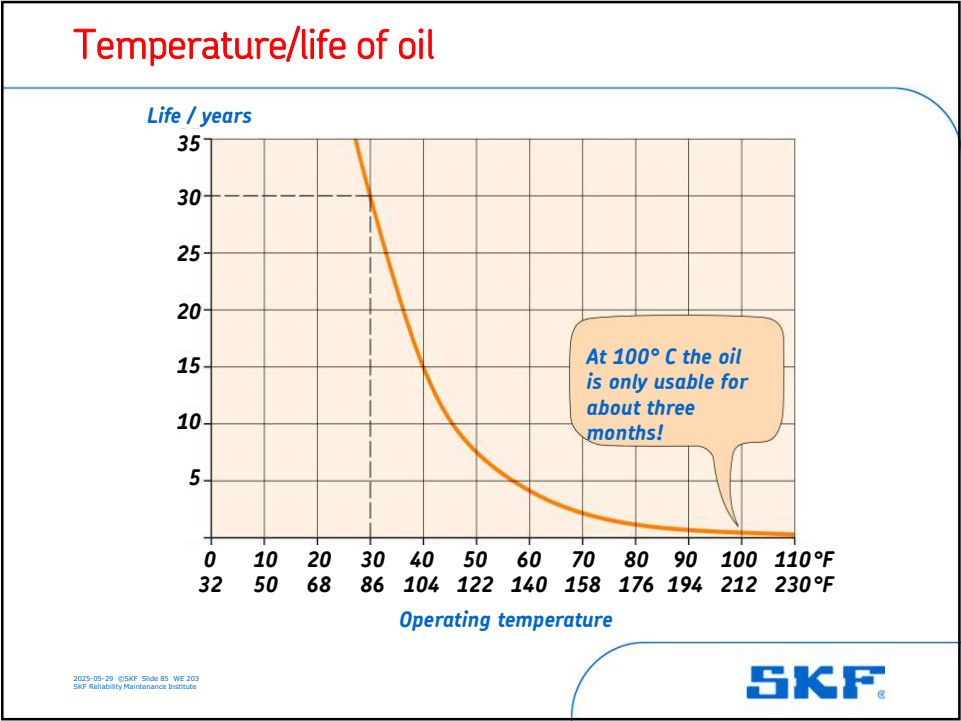
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Grease or oil

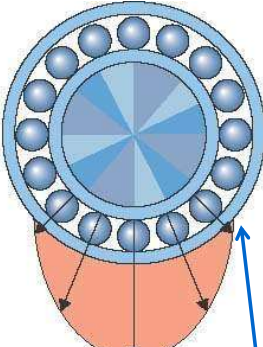
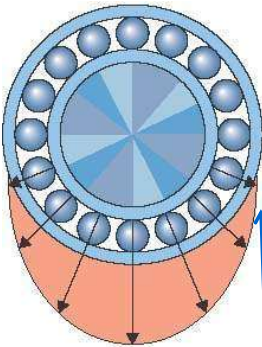
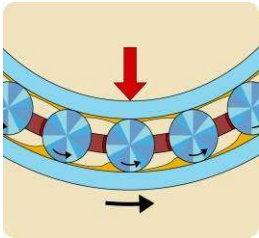


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Oil: minimum load issue



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Oil: cage design concern



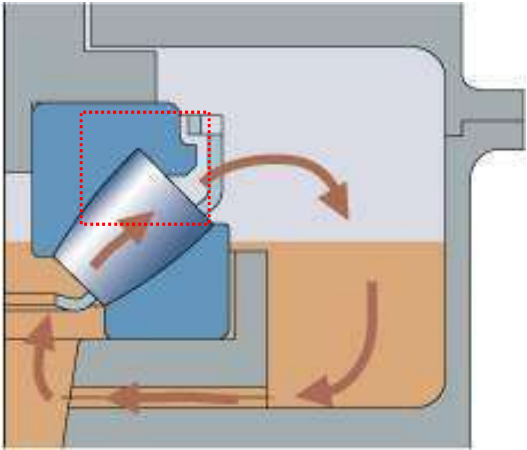
A close-up photograph of a bearing cage, showing the internal structure and the SKF logo engraved on the outer ring.

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Oil: bearing design



The left image is a 3D model of a bearing cage. The right image is a cross-section diagram of a bearing cage showing the internal oil flow. The diagram illustrates the oil flow path, with arrows indicating the direction of flow. A red dashed box highlights the area where the oil flow is most critical.

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Grease: no complicated sealing system



A line drawing of a worker wearing a hard hat and a suit, applying grease from a tube into a bearing housing. A large blue puddle of grease is shown leaking out from the bottom of the housing, illustrating the lack of a complex sealing system.

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Grease: no special housing design



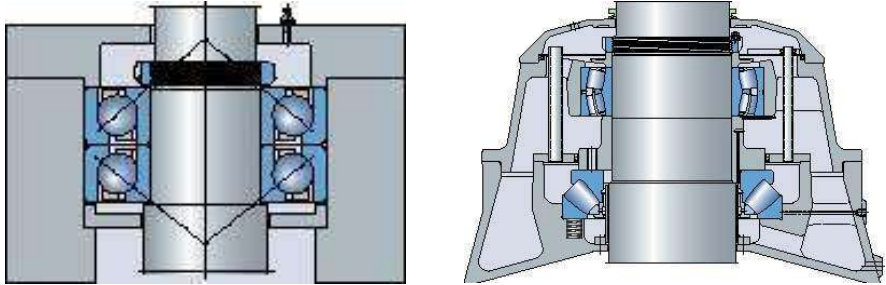
A 3D cutaway view of a bearing housing. It shows a standard design with a central bearing cavity and a grease nipple at the bottom. The design is simple and does not require special features for grease lubrication.

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Grease: vertical shafts

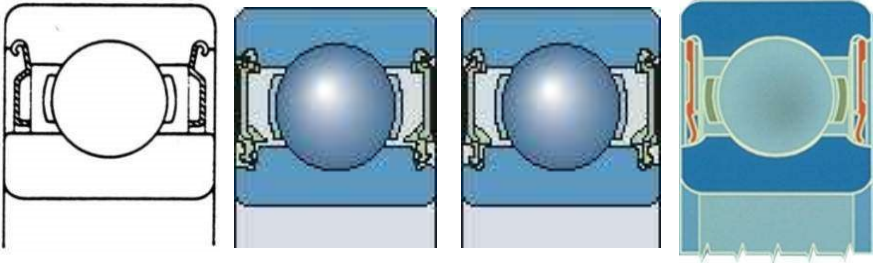


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Grease: maintenance-free designs



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Grease: easy



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